MINOR PROJECT – 2

PROJECT INTRODUCTION

Project Title

Predict New York City Airbnb Prices

Name

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Dataset Name

New York City Airbnb Open Data (AB_NYC_2019.csv)

Dataset Link

https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data

Short Description of the Dataset

The dataset contains 48,895 Airbnb listing records in New York City and 16 columns describing each listing. It includes listing and host identifiers, listing name, neighbourhood and neighbourhood group, latitude and longitude, room type, price, minimum nights, number of reviews, most recent review date, reviews per month, calculated host listing count, and availability over 365 days. These features can be explored to understand how location, room type, reviews, and availability relate to listing prices.

Objective of the Analysis

The objective of this project is to build a Linear Regression model to predict Airbnb listing prices in New York City. The work includes exploring the dataset structure and data types, identifying and handling missing values, removing unnecessary columns, examining and handling outliers, encoding categorical features, splitting the data into training and testing sets, training the model, and evaluating its performance using R^2, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). Actual-versus-predicted and residual plots will be used to assess prediction quality and identify whether the model performs differently for cheaper and more expensive listings.